

## RAG Comparison: FAISS + OpenAI vs ChromaDB + HuggingFace

### Introduction

This experiment seeks to evaluate the performance of two Retrieval-Augmented Generation (RAG) pipelines with varying vector databases and embedding models. The performance will be based on the accuracy of the response, speed of processing, and complexity of the setup for retrieving data from a dataset related to renewable energy.

### Dataset

To conduct the experiment, the dataset was created based on the Wikipedia webpage regarding the topic "Renewable Energy." The webpage was loaded into the LangChain's WebBaseLoader and then segmented into smaller chunks with the use of the Recursive Character Text Splitter. Then, the chunks were converted into vector embeddings.

### Test Query

In order to evaluate the RAG pipelines, the same query has been used for both systems:

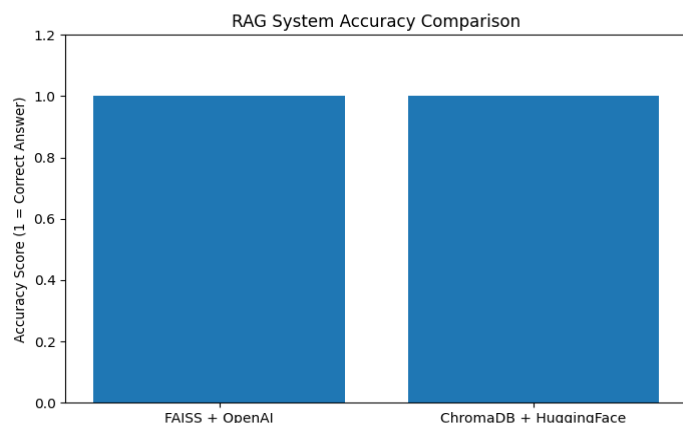
"What are the main types of renewable energy sources?"

This query has been chosen because it involves the retrieval of factual information from the renewable energy dataset. By using the same query for both pipelines, it has become easy to compare the response time.

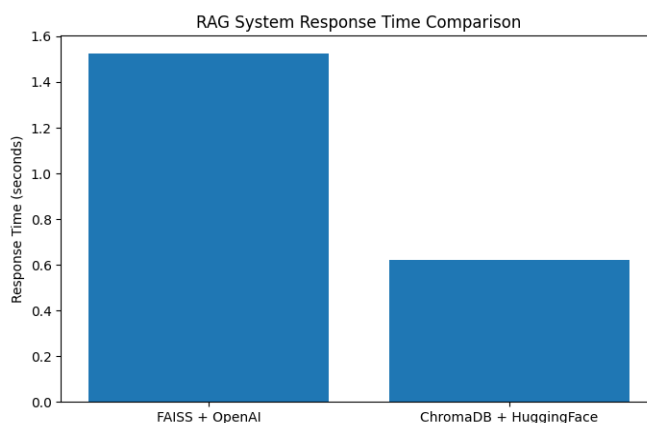
### Results

#### Response Accuracy

The RAG systems were able to retrieve the context from the dataset and give the right answers. The responses included the important types of renewable energy, which are solar power, wind power, hydroelectric, geothermal, and biomass. (Refer Fig 1.)



**Fig 1.** Accuracy comparison between FAISS + OpenAI and ChromaDB + HuggingFace. Both systems achieved the same accuracy for the test query.



**Fig 2.** Response time comparison between the two RAG systems. ChromaDB + HuggingFace responded faster in this experiment

#### Processing Speed

For measuring the response time of the systems, the "time" module of the Python program was used. The response time for the Chroma DB pipeline was faster than that of the FAISS pipeline. (Refer Fig 2.)

#### Setup Complexity

| System                 | Setup Complexity |
|------------------------|------------------|
| FAISS + OpenAI         | Easy             |
| ChromaDB + HuggingFace | Medium           |

### Observation

Both RAG systems were able to produce accurate answers using the same dataset. Both systems were able to generate accurate results using the same data set. The response time of the ChromaDB + HuggingFace system was faster than that of the FAISS + OpenAI system.

### Pros / Cons Comparison

#### FAISS + OpenAI

Pros: Easy integration, high-quality embeddings.  
Cons: Requires OpenAI API and may incur cost.

#### ChromaDB + HuggingFace

Pros: Open-source embeddings, faster response time.  
Cons: Slightly more setup and dependency installation.

### Conclusion

Both RAG pipelines were successful in retrieving the relevant information and generating the accurate responses. However, the system to be used would depend on the project needs, for instance, the cost, the speed, and the complexity of the implementation. Therefore, the use of the FAISS system with the Open AI embeddings would offer a simpler implementation and the use of the Chroma DB system with the Hugging Face embeddings would offer a faster response and low dependency on the external API.